Model Report Phase 2

IGC Pharma, Inc

Abstract

Early detection of Alzheimer's disease and related dementias (ADRD) is essential for timely intervention and slowing symptom progression. Speech-based diagnostic tools have emerged as promising biomarkers due to their non-invasive, cost-effective nature and their ability to reveal early cognitive decline through advanced Machine Learning (ML) and Deep Learning (DL) techniques [1]. In this work, we fine-tuned seven models across different modalities, including Whisper, a model primarily designed for speech-to-text conversion[2]. Among these, Whisper demonstrated the best performance. By leveraging its encoder representations in a multilingual context, we extracted robust audio features and employed linear layers for the classification task. This approach achieved a log loss of 0.6976 on the private leaderboard, highlighting its effectiveness. Whisper's extensive pretraining on diverse multilingual corpora enabled us to address the dataset's linguistic diversity effectively. This work provides a practical and scalable approach to early ADRD detection, integrating state-of-the-art DL techniques and demonstrating the potential of language-inclusive diagnostic tools.

Methodology

Data Preprocessing

The audio inspection showed consistent background noise, loudness variability, and poor intelligibility. To overcome the audio quality challenges, we apply a voice enhancement preprocessing pipeline that implements stationary noise gate reduction, dynamic noise gate and equalization, dynamic range processing, and loudness normalization. Firstly, we used the noisereduce [3] python package, which analyzes a sample of pure background noise from the audio to create a noise profile. The entire audio signal is processed using the noise profile and, through the Fourier transform, identifies and removes similar noise patterns throughout the recording. It creates and smooths a mask that effectively filters out the background noise while preserving the desired speech components. This process results in cleaner audio where the speech is more prominent, and the constant background noise interference is minimized. Dynamic noise gates and equalization were used to attenuate residual noise during speech pauses. We applied a dynamic noise gate with a -30dB threshold, 1.5 ratio, and 250ms release time. Then, we equalized the audio signal using a series of filters: an 18kHz lowpass, and a 100Hz high-pass filter. Three peak filters at 250Hz (1dB gain), 500Hz (-1dB gain), and 2kHz (1.5dB gain, Q=2.5) were used to enhance vocal presence and clarity. These frequencies were selected according to the frequency range of the human voice.

We employed a compressor with a -12dB threshold and 3:1 ratio during dynamic range processing to address amplitude variability, reducing the speech signal's dynamic range. This stage helps maintain consistent speech levels while preserving natural vocal dynamics. Finally, we implemented the ITU-R BS.1770-4 loudness normalization algorithm for loudness normalization. The algorithm measures and adjusts audio loudness using Loudness Units Full Scale (LUFS). First, it applies K-frequency weighting to match the human perception of loudness at different frequencies, followed by mean square calculation applied to overlapping windows and gating blocks to exclude low-intensity audio signal segments. The audio is normalized to our target of -14.0 LUFS (a standard for digital streaming platforms), which ensures consistent perceived loudness across all processed audio files while preserving the relative dynamics within each recording.

Audios were processed using the pipeline and then reviewed to remove clips where the interviewer talked predominantly but were labeled as MCI or AD. We removed unintelligible audio tasks due to noise or poor audio quality consisting of twenty-four audios from English tasks and five from Spanish tasks.

Data Augmentation

Data augmentation is used in DL models to generalize capabilities and reduce overfitting. This technique expands the training dataset by creating modified versions of the input data while preserving the underlying content and labels. We implemented an audio augmentation strategy using the torch-audiomentations library. We sequentially applied three transformations, each with a 30% application probability to any given sample. The first transformation synthesizes and introduces colored noise to the audio signal. Then, we applied random pitch modifications within a range of ±3 semitones while preserving other signal characteristics. This transformation helps the model become invariant to natural pitch variations due to differences in speaker characteristics or recording conditions. The final transformation implements dynamic range augmentation by applying random gain adjustments between -6 and 6 decibels. This technique simulates the variation in recording volumes and microphone distances commonly encountered in real-world scenarios.

Tasks Analysis

Transcriptions from the selected raw audios were extracted with OpenAI's whisper models [4]. The medium model provided the best transcriptions while keeping a good inference speed. The model automatically finds the language of the audio and transcribes or translates the spoken words to English. This processing allowed us to identify the distribution between languages, tasks, and diagnoses. English was the most predominant from the raw audios, followed by Spanish and Mandarin. We identified four different tasks from the biggest to smallest samples in the data: image description (English), Alexa instructions (English), reading fragment from Don Quijote de la Mancha (Spanish), and animal naming (Mandarin). Table 1 presents a summary of the analysis conducted for each task. This new metadata allows for stratification in training and validation subsets. With more data, we could develop specific models for each task. Please note that in the case of the Alexa task, participants didn't propose their own questions but instead read them from a shared list, implying that the Alexa task evaluates domains similar to the main reading task but with lower complexity.

Table 1. Tasks descriptions.

Task	Domains assessed	Importance
Image task	• Expressive language (the ability to articulate thoughts into coherent sentences, engaging grammar, vocabulary, and syntax) • Fluency (continuous, uninterrupted speech) • Semantic processing/impairment (involves naming objects, identifying relationships, and integrating visual stimuli into a meaningful narrative.)	• It taps into the ability to produce connected discourse, reflecting deficits in language or memory that are often early indicators of MCI or AD. • Picture description has been widely validated in neuropsychological research for assessing narrative coherence and idea density, and both are impaired in MCI and dementia.
Animal task	• Verbal Fluency (the generation of words within a semantic category (animals) under time constraints) • Processing speed (the efficiency of lexical retrieval and rapid cognitive associations)	• This word-generation test asks patients to name as many four-legged animals as they can. People with dementia are 25 times more likely to name fewer than 10 animals in one minute (Rofes et al., 2020). • Animal listing is a gold standard in neuropsychological assessment for identifying impairments in semantic memory, category clustering, and switching. These deficits are characteristic of MCI and AD. • Processing speed is an auxiliary factor, as slower retrieval often correlates with cognitive decline.
Reading task	• Reading Fluency (the ability to read text aloud smoothly and with correct intonation) • Phonological Processing (the ability to decode words, articulate phonemes, and maintain accuracy in pronunciation – acoustic abnormality) • The reading task is based on the first paragraph (126 syllables) of the novel by	• Dysfluencies, mispronunciations, and hesitation during reading are often observed in neurodegenerative conditions, reflecting underlying deficits in language processing or working memory. • While not typically used as a standalone task for dementia screening, reading passages can complement other measures by highlighting phonological errors and prosodic changes.

	Miguel de Cervantes, *Don Quijote de la Mancha*.	
Alexa task	• Speech production and clarity: the ability to articulate commands accurately that Alexa could recognize properly	• The task provides a fixed set of commands to read, making it a comparable task across participants.

Modeling

We established a series of baseline models to design the experimental methodology for the challenge, progressing from traditional to more recent AI techniques. This progression began with traditional ML approaches and advanced to DL models, ultimately incorporating transformer-based architectures. Regarding ML, the inputs for experimentation were the pre-generated acoustic features provided in the challenge. We performed model sweeps through PyCaret to determine the best architectures. Those sweeps evaluate base models from which we selected the top 3 performing architectures: Random Forest, XGBoost, and CatBoost. We trained the three models using cross-validation techniques to tune their parameters at a more granular level.

After evaluating ML baselines, experiments moved to DL architectures, starting with Recurrent Neural Networks (RNNs). From this point forward, the raw audios were used as input instead of the features provided. The first models tested involved LSTM layers followed by a classification linear layer both with and without adding age and gender metadata. Considering the variability of audio length across the data, two sets of models were trained: padded and segmented audio models. The former involved padding and, in some cases, trimming audio to 30 seconds, while the latter consisted of splitting the audio into 0.2-second segments and aggregating all predictions from segments belonging to the same audio to provide a final prediction.

As was pointed out by Yang et al. [5] and Ding et al. [1] in their literature review on speech-based AD detection, pre-trained models consistently achieve higher accuracy compared to traditional ML approaches and non-trained DL models. Therefore, we focused our next set of experiments following a transfer learning approach with self-supervised transformer-based architectures like Audio Spectrogram Transformer (AST), HuBERT, and Whisper. The AST converts audio inputs into spectrograms using a Mel filter bank over 10ms windows [6]. Those spectrograms go through linear projections, a transformer encoder, and a final linear layer. We focused experiments on optimizing the learning rate, mix-up ratio between classes, and masking percentage. This model is interesting because of its transfer learning capability based on images and video from larger datasets.

In contrast, HuBERT [7] (Hidden-Unit BERT) takes a different approach to self-supervised learning, focusing specifically on speech representation. The pre-training of the model first clusters the acoustic features to create pseudo-labels. Then, these clusters serve as targets for a masked prediction task, like BERT's masked token prediction. We performed several experiments with HuBERT using the large and small models and the CLS token as input for a classification head. Additionally, we performed experiments following a multimodal approach, using the text transcriptions and the raw audios as inputs to the HuBERT and BERT models. We extracted the CLS token from both models and explored different fusion techniques such as concatenation, additive fusion, and attention-gating mechanisms.

Finally, we explored OpenAI's Whisper model [4], a state-of-the-art multilingual speech recognition system. We selected this model for its ability to process up to 97 languages, which enables generalizability and robustness regarding languages. It is essential to highlight that 82.7% of training data was in English. However, the other 17.3% corresponds to 117,000 hours of audio in 96 languages, of which some, such as Spanish, Catalan, Afrikaans, and Swahili, are spoken in AD prevalent communities. It is primarily designed for speech-to-text conversion, but we leveraged its encoder representations for our classification task. We found it interesting to evaluate if a multilingual model improves the performance, considering the three different languages of the dataset.

Performance analysis

Table 2. Summary of Experimental Results. DP – Default Parameters; OP – Optimized Parameters; PA - Processed audios; MD - Age & Gender metadata; AU -Audio augmentations; Sub – Submission number at the leaderboard.

Model	**Specifics training details**	**Private log loss**	**Validation log loss**	**Train log loss**
Random Forest	DP (Sub. 2)	0.8128	0.6419	
	OP 4-fold (Sub. 5)	0.8175	0.8011	
	DP all data (Sub. 9)	0.8108	-	
	DP all data aggregated segments (sub. 19)	0.8068	-	
CatBoost	Optimized 4-fold (sub. 6)	0.8374	0.8227	
	Default Parameters all data (sub. 11)	0.8340	-	
XGBoost	Optimized 4-fold (sub. 7)	0.8373	0.8257	
	Default Parameters all data (sub. 13)	0.8377	-	
LSTM	2 stack + linear layer (sub. 3)	0.9293	0.7127	
Audio Spectrogram Transformer	AST scratch (sub. 14)	1.0103	0.9809	
	AST fine-tuned (sub. 15)	0.9613	1.0136	
	AST by segments with PA (sub. 21)	0.8661	0.9736	
	AST by segments with MD (sub. 24)	0.8362	0.9047	
HuBERT	HuBERT Large with AU	1.0973	0.9960	0.6869
	Base + AU	-	0.7427	0.7391
	HuBERT Base with AU and BERT (sub. 33)	0.8087	0.8146	0.7337
	HuBERT Base with AU and MD (sub. 34)	0.8620	0.8527	0.8473
Whisper	Whisper Base		0.7271	0.6385
	Whisper + RoBERTa		0.7137	0.6685
	Final Whisper (sub. 41)	0.7191	0.6963	0.7257
	Final Whisper, 1 more epoch (sub. 42)	0.6976	-	-
	Final Whisper, 2 more epochs (sub. 43)	0.6976	-	0.6680

Traditional ML models using the provided acoustic features achieved moderate but consistent performance, with Random Forest showing the most stable results (private log loss 0.8108-0.8128). While these models offer reliable baselines, their performance suggests limitations in capturing complex temporal speech patterns characteristic of cognitive decline. The transition to deep learning architectures initially showed mixed results. Basic LSTM models underperformed (0.9293 private log loss), likely due to the limited dataset size and the challenge of learning robust speech representations from scratch. The Audio Spectrogram Transformer (AST) experiments showed interesting patterns. While initial attempts at training from scratch yielded high log losses (1.0103), fine-tuning from the complete Audio Set database pre-trained model combined with audio preprocessing substantially improved performance (0.8362). Notably, AST demonstrated unique behavior regarding demographic features such as age and gender, which improved performance (from 0.8661 to 0.8362), contrasting with other transfer learning approaches where demographic data decreased performance.

The HuBERT experiments exhibited significant performance improvement. While the base model achieved a validation loss of 0.7638, the large model showed signs of overfitting with our internal train-validation loss disparity (0.6869 vs 0.9960). Data augmentation strategies improved model stability (our validation loss 0.7427), though adding demographic features degraded performance (0.8527). Our multimodal fusion approach combined HuBERT and BERT through attention-gating mechanisms and achieved similar results (0.8146).

The most significant performance improvements came from the fine-tuned Whisper model, achieving a private log loss of 0.6976. This substantial improvement over other architectures can be attributed to Whisper's robust multilingual pre-training and effective transfer of speech understanding capabilities. The model's consistent performance across multiple submissions (0.6976-0.7191) suggests stable learning and better generalization. However, similar to HuBERT, adding demographic features slightly decreased performance (0.7137). For our final submission, we used the remaining data as a validation set to train the whisper model for two more epochs, yielding the best log loss value on both the public and private test sets.

Bias and ethics analysis

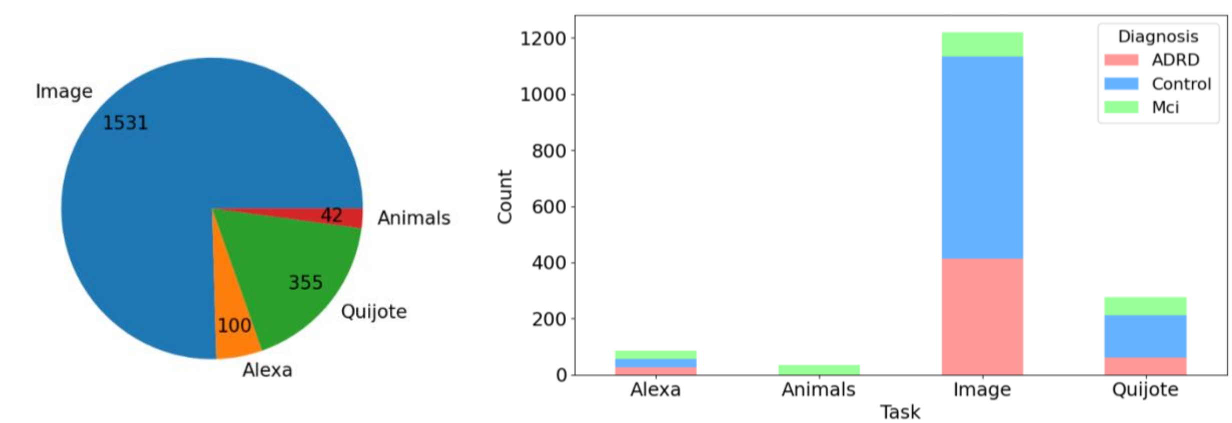
The data provided for training presented an issue for generalizability as tasks weren't balanced across diagnosis and languages while assessing different domains. Figs. 1A and 1B show that most data correspond to English in the image description task with an uneven diagnosis distribution, particularly for MCI subjects. Additionally, for the animal naming task, which was only conducted in Mandarin, there were only MCI samples. This created a challenge in training AI models as they extracted information from the available data. In this case, the data distribution for training may introduce strong biases towards language and predominant diagnoses. For once, AI models may identify that every time there is an audio in Mandarin, it will be MCI because there are no examples of Mandarin with AD or control diagnosis.

Furthermore, the gender distribution is slightly unbalanced, with 42% of the subjects being male in the train split and 37% in the test. Fig. 2B illustrates the model's log loss by gender in the test set, which shows a significant decrease in performance between the male and female subjects, with an average log loss of 0.771 and 0.654, respectively.

The age-stratified analysis in Fig. 2A shows that the model peak performance for female subjects is for the age group of 55-59 and males 65-69, with a clear deterioration pattern of performance as the age increases; we argue that speech clarity could deteriorate as normal aging occurs, and that for younger subjects it is more difficult for the model to detect speech patterns related to cognitive decline and ADRD as the disease is less evident. However, the sample size for the age groups is unbalanced and could bias the model's prediction to the more prominent groups.

Many strategies can be used to address the bias from the dataset in the challenge. The most straightforward option is to look for additional data to increase the samples in the underrepresented groups. However, real-world data with balanced distributions are not common. Then, from the algorithm side, data can become balanced with over and under-sampling techniques that have been proven effective in increasing performance and reducing the bias of AI models to majority classes as they level the distributions of the training classes [8].

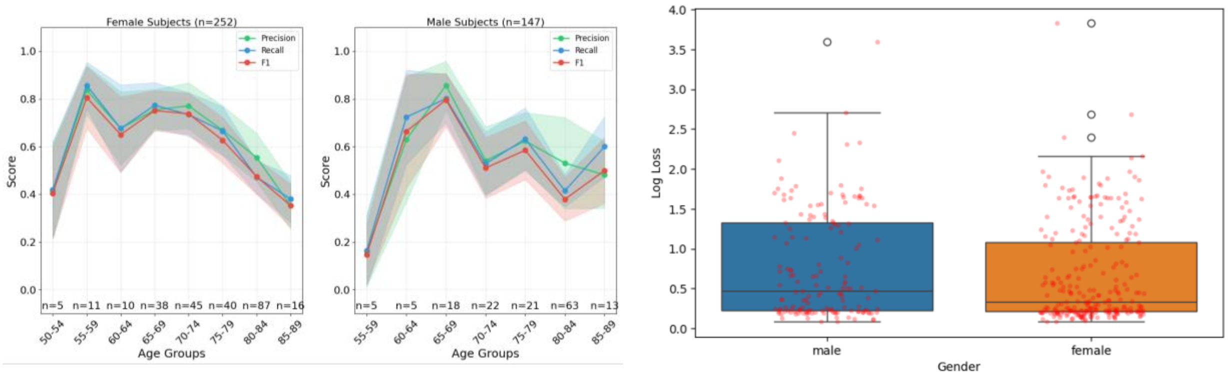
Regarding privacy in speech and audio-related datasets, voice cloning techniques can be applied to anonymize recordings and protect participants from being identifiable later. However, removing participants' metadata from training significantly reduces the risk of identification. These metadata include the site where their data was collected, contact details, or even data from healthcare professionals who interacted with the participant.



A. All dataset tasks

B. Train subset tasks by diagnosis

Fig. 1. Data distribution.



A. Performance metrics by gender and age group B. Log loss distribution by gender

Fig. 2 Model performance by age and gender

Error analysis

The confusion matrices in Fig. 3 illustrate how accurately the best-performing model classified the test subset for each task. As previously mentioned, issues related to data bias are also evident during model inference. In the image task, key challenges arise with MCI, which has a very small sample, and ADRD, which shows a high degree of misclassification as control participants. The overlapping cognitive features across stages of decline likely contribute to these misclassifications. For instance, the image task reveals confusion between these groups and controls, highlighting the diverse ways cognitive decline impacts expressive language and semantic processing. Tasks requiring visual integration, narrative articulation, and memory usage demonstrate how deficits evolve and overlap across stages of deterioration, complicating the identification of clear diagnostic boundaries [9,10]. Clinically, this underscores the continuum of neurodegenerative decline, where MCI often presents subtle impairments resembling normal aging or early ADRD. Differentiating based solely on cognitive performance is challenging, as individuals may display similar deficits or compensatory mechanisms [11]. For example, an MCI participant with preserved semantic abilities might perform like a control, while an early ADRD participant with mild memory loss could resemble MCI.

In the Alexa task, the classification bias is evident as most participants were classified as MCI, regardless of their diagnosis. This could be due to group selection biases, such as older adults familiar with voice assistant systems demonstrating higher clarity in commands. However, this task's low complexity and lack of spontaneous speech features limit its clinical relevance [12]. While structured speech tasks, as suggested by Liang et al.[13], improve standardization by using a fixed set of questions, this approach sacrifices the nuanced spontaneous speech patterns associated with cognitive decline, such as alterations in discourse planning, hesitations, or semantic errors. The Quijote reading task further emphasizes the challenge of limited data representation. Misclassifications of MCI as control and ADRD participants suggest that the task, while helpful in detecting phonological and prosodic changes, lacks sufficient complexity or sensitivity to distinguish intermediate stages of cognitive decline. This aligns with the observation that reading tasks are supplementary in dementia screening, as they may not fully capture higher-order language and memory deficits unless paired with other diagnostic measures [14].

Finally, despite achieving accurate classification for MCI, the animal listing task lacks data for control and ADRD groups. While verbal fluency tasks like this are gold standards in neuropsychological assessments, the absence of diversity in the dataset limits the model's ability to generalize across diagnoses. Semantic fluency deficits in ADRD often differ from MCI regarding clustering and switching strategies, which this model could not assess due to limited data [15].

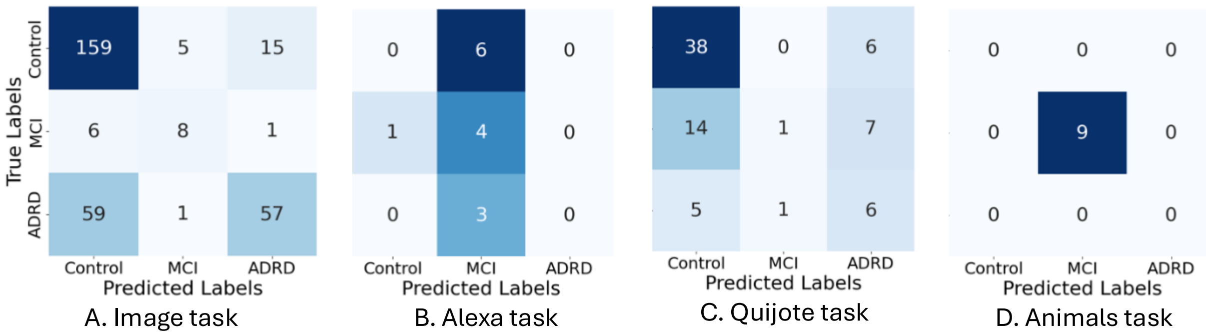


Fig. 3. Confusion Matrix for each task.

Overall, the biases observed highlight the clinical complexity of distinguishing cognitive states like MCI and ADRD. The overlapping features, limited data representation, and lack of task diversity reflect real-world diagnostic challenges in dementia care. Furthermore, while the Montreal Cognitive Assessment (MoCA) is a widely used screening tool chosen in some studies' databases from this challenge, its standalone use is insufficient for a clear and accurate diagnosis. Sensitivity and specificity may vary across cultural contexts and educational levels, potentially leading to misclassifications and affecting diagnostic cutoffs [16].

Future directions

The unexpected impact of demographic features across models reveals significant opportunities to address model limitations and improve performance in and beyond Phase 3. While metadata such as age and gender improved AST performance, it degraded results in HuBERT and Whisper models. The counterintuitive outcome suggests limitations in current feature integration techniques. In future work we will explore two main approaches to address these challenges.

First, we will use advanced integration methods, such as learned embeddings or feature specific attention mechanisms, to better leverage demographic factors critical to Alzheimer's diagnosis. Additionally, although multimodal approaches (combining audios and text) showed promising initial results, the superior performance of fine-tuned single-modality models show the need for advanced fusion techniques. In Phase 3, we plan to use hierarchical multimodal attention networks or cross-modal contrastive learning, to better exploit complementary information from different modalities while preserving the benefits of transfer learning.

Second, we will address limitations in Whisper's architecture by modifying its source code. Masked attention, widely used in sequence-based models in domains for language[17], audio [18], and vision[19] offers advantages, such as handling variable length input, next token prediction tasks, and missing information. Whisper models, however, lack masked attention, which complicates handling variable length audios. In Phase 3, we also plan to modify Whisper's architecture to incorporate masked attention. We hypothesize that by combining its pretrained weights in multiple languages with the enhancement of masked attention will improve Whisper's performance.

We plan to apply the advanced fusion techniques after the enhanced Whisper model processes audio data, the strengths of both methods will be combined. The Whisper model's improved audio representations—more robust due to masked attention—will integrate with text data and demographic features through the fusion framework. We hypothesize this integration will allow the model to leverage complementary information from all sources while maintaining accuracy and consistency. The combined approach ensures that each modality (audio, text, and metadata) contributes its unique strengths. We believe that together, these techniques will enhance the overall model's ability to detect Alzheimer and MCI with greater accuracy.

Finally, mitigating bias is a key focus. At the pre-processing stage, we will implement demographic sampling strategies to address imbalances. For in-processing, we expect to explore adversarial learning techniques, incorporate fairness constraints into loss functions, and apply regularization penalties to discourage biased predictions[20]. These measures are expected to ensure equitable and robust model performance across diverse demographic groups.

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